

Version 7 – Plot Guide

How shear strain changes a crystal's vibrations, and where it breaks bonds – what every plot shows, where its data comes from, what physics it tells, and what to expect in real life.

The short story. We shear a crystal in small steps. At every step we ask two questions: *how do the vibration frequencies (phonons) change?* and *which bonds are overstretched and break?* Six plots (P1–P6) answer the first, two plots (Q1–Q2) answer the second. Everything is built from the crystal your app has loaded (built-in or any CIF), so the results always match the model you are looking at.

Contents

1. How the analysis works (data flow)
2. The three model levels
3. Phonon plots P1–P6
4. Bond-breaking plots Q1–Q2
5. Assumptions at a glance
6. Model vs real crystals · 7. How to read results honestly

1. How the analysis works

1.1 What comes from your existing code

Quantity	Produced by	Meaning
Unit cell and atoms	<code>parseCIFFile.m</code> , <code>getBuiltinCrystalDef.m</code>	lattice vectors, fractional positions, atomic masses
Atom positions <code>pos</code>	<code>generateLatticeGeneral.m</code> (called by <code>autoRigidShells.m</code>)	$N \times N \times N$ supercell, one row per atom
Bond list <code>bonds</code>	<code>autoRigidShells.m</code> , <code>findPeriodicNeighborCandidates.m</code>	rows $[i \ j \ d_0 \ u_x \ u_y \ u_z]$: two atoms, rest length, unit direction. One spring per neighbour image
Masses, spring constant k	CIF / app field	one mass per atom, one k for every bond
Shear convention	<code>applyShearForce.m</code> , <code>computeShearForces.m</code>	$u_x = \gamma \cdot z$; bond vectors are rotated as $\mathbf{r} = \mathbf{r}_0 + \gamma \cdot \mathbf{r}_0 \times \hat{x}$ (safe for periodic wrap-around)

1.2 What is computed at every strain step (`phononShearSweep.m`)

1. **Bond geometry:** new bond vector, length and stretch $(d - d_0)/d_0$.
2. **Bond law** (`shearBondModel.m`): force and local stiffness from a Hooke or Morse spring.
3. **Breaking:** a Morse bond past its force maximum is removed permanently.
4. **Stiffness matrix $\mathbf{K}$** (`assembleShearHessian.m`): each bond adds a 3×3 block; optional “pre-stress” term for stretched bonds.
5. **Dynamical matrix $\mathbf{D} = \mathbf{M}^{-1/2} \mathbf{K} \mathbf{M}^{-1/2}$** ; the three rigid translations are projected out.
6. **Eigenvalues $\lambda = \omega^2$** (negative means unstable) and the six lowest eigenvectors (`eig`, or your `jacobiEigenSolver.m`; they agree to 2×10^{-14}).
7. **Shear stress $\tau_{xz} = (1/\text{Vol}) \sum \mathbf{f} \cdot \mathbf{r}_x \mathbf{r}_z / d$** over bonds (virial formula).

8. **Bookkeeping:** which bonds broke and when, first strain with $\omega^2 < 0$, non-affine relaxation residual, and the “soft band” frequency.

Units: ω is in $\sqrt{(k/m)}$ with your k and the CIF masses; stress in k/length^2 ; strain has no units.

Stable range. All phonon plots stop at the last strain before the first $\omega^2 < 0$. Beyond that the relaxed structure has moved to a different configuration and the numbers no longer describe your crystal.

2. The three model levels

#	Name	Bond law	Pre-stress	Relaxation	Breaking	Why it is there
1	harmonic, affine	Hooke, stiffness k	no	no	no	Exactly your sliders / <code>applyShearForce.m</code> (checked to 10^{-15}). Softening only from bond rotation.
2	harmonic + pre-stress + relaxation	Hooke	yes	yes	no	What equilibrium relaxation and stretched-bond stiffness add.
3	Morse + pre-stress + relaxation + rupture	Morse, $a = 4/d_0$	yes	yes	yes	Main model: the only one that can fail.

Morse bond: $V = D(1 - e^{-a\delta})^2$ with $D = k/(2a^2)$, so the stiffness at rest is your k . Stiffness falls as the bond stretches and reaches zero at the force maximum $\delta_c = \ln 2/a$ – that is where the bond “breaks”. With $a = 4/d_0$ every bond breaks at the same relative stretch, $\ln 2/4 = \mathbf{17.3\%}$.

Relaxation: after the affine shear, atoms in a multi-atom cell are not in force balance. We solve for the periodic internal displacement that restores balance (Newton with line search, warm-started from the previous strain; leftover force $\sim 10^{-9}$).

3. Phonon plots P1–P6

P1 Softening curve

How much lower do the low-frequency vibrations get as the crystal is sheared?

What you see. Left: soft-band frequency and mean frequency of all modes, divided by their unstrained value (Morse model). Right: the soft band for all three model levels. Orange dotted line = first bond breaks; red dashed line = first negative ω^2 .

Data in. `omegaSoft` and eigenvalues from the sweep (built from **your** pos/bonds/masses/k); event strains from the bond-break and instability records.

How it is made. At each strain, diagonalise D , average the lowest 5 % of eigenvalues (at least 3), take the square root, divide by the $\gamma = 0$ value.

Physics. Two effects lower frequencies: bonds rotate (this alone gives $\omega \approx 1 - c\gamma^2$, the blue curve) and bonds stretch into the anharmonic region and break. The curve hitting zero is the **elastic (Born) instability**: the lattice can shear spontaneously.

Real life. A few percent of shear changes phonon frequencies by well under 1 % and mainly splits degenerate modes (first-order shift vanishes by symmetry; second order $\propto \gamma^2$). Big softening appears only near the ideal shear strength, roughly $\gamma \approx 0.1$ – 0.3 . Real crystals seldom reach it because defects relieve the strain earlier.

Assumes. Morse $\alpha = 4$; one k for all springs; soft band = lowest 5 % (chosen because the single lowest mode of a barely rigid network, e.g. Al_2O_3 with $\lambda_{\min} = 8 \times 10^{-4}$ against ~ 0.6 at the top, jumps around); Gamma-point modes of the chosen supercell only.

Caveat: the pre-stress model can *raise* the lowest modes in low-symmetry crystals (tension in springs stiffens nearly-floppy directions, like a tensegrity). That is a spring-network effect, not a contradiction.

P2 Density-of-states shift

Where do the vibrational states sit before and after shear?

What you see. Phonon DOS $g(\omega)$ at $\gamma = 0$ (blue) and at the last stable strain (orange), dashed lines at each mean frequency; the title gives the % shift of the mean and of the soft band.

Data in. All eigenvalues at those two strains.

How it is made. Each mode ω_n becomes a Gaussian of width 1.5 % of the top frequency; the Gaussians are summed on a 700-point grid. Area = number of modes.

Physics. The DOS controls low-temperature heat capacity, vibrational entropy and the number of phonon states an electron can scatter into. Softening moves weight to lower ω .

Real life. Tenths of a percent shift for a few-percent shear, with the low-frequency edge moving most. Near failure, states pile up at low ω – the classic precursor of a structural transition.

Assumes. Broadening is cosmetic. Only supercell-folded Gamma modes exist, so a small cell gives a spiky DOS (a 30-atom cell has 87 modes); it is not a converged k-space DOS.

P3 Spectral map

The whole spectrum versus strain in one picture.

What you see. Heat-map: strain on x, frequency on y, darkness = DOS. Each dark ridge is a phonon band; the red line is the lowest mode.

Data in. Eigenvalues at every stable strain step.

How it is made. The P2 broadened DOS is computed for every strain and stacked as columns (`imagesc`).

Physics. Ridges bending down = softening; ridges that repel each other = mode mixing; ridges rising = stiffening; the red line reaching the floor marks the instability.

Real life. Bands drift smoothly and typically *one* branch (a transverse acoustic or shear-like one) falls to zero while the rest barely move. Sudden jumps are not expected in a real crystal; here they appear only where bonds are removed.

Assumes. Same as P2.

P4 Softest-mode atoms

Which atoms actually move in the mode that is about to fail?

What you see. Two panels ($\gamma = 0$ and last stable strain). Each atom is a marker on the xz projection of the sheared cell; size and colour = normalised displacement $|u_i|$; arrows = in-plane displacement direction.

Data in. Lowest eigenvector from the sweep; `extractModeShape.m` (yours) converts it to real displacements $u = M^{-1/2}v$ and scales the largest atom to 1; atom positions = pos + $\gamma \cdot z$ + relaxation displacement.

How it is made. The $[N_{\text{atoms}} \times 3]$ output gives $|u_i|$ (row norms) for colour/size and its x, z columns for arrows. Atoms are drawn in ascending $|u|$ so the most mobile sit on top.

Physics. A *localised* pattern (a few big markers) points to a nucleation site; a *coherent, uniform* pattern points to a collective shear instability. Comparing panels shows how the pattern reorganises near failure.

Real life. Pristine crystals: an extended shear wave. Localisation is expected next to vacancies, dislocation cores and surfaces – that is where plasticity begins.

Assumes. Only the single lowest eigenvector (an arbitrary mix if modes are degenerate); xz projection overlaps atoms that differ only in y; only patterns that fit the supercell can appear.

P5 Stress–strain vs phonon softening

Mechanics and vibrations tell the same story.

What you see. Left axis (blue): shear stress $\tau_{xz}(\gamma)$. Right axis: soft-band ratio (orange) and shear-wave speed ratio $\sqrt{G_t/G_0}$ (purple dashed).

Data in. `tau` from the sweep, using the same virial formula as `computeAtomicStress.m` (checked against `computeShearForces.m`); `centralDiffDerivative.m` (yours) applied to a smooth interpolant of $\tau(\gamma)$ gives the tangent modulus $G_t = d\tau/d\gamma$; soft band from the sweep.

How it is made. Stress rises linearly (slope G_0), bends as bonds soften, peaks at the ideal shear strength and drops when bonds break. G_t falls to zero at the peak. $\sqrt{G_t/G_0}$ is the relative speed of long-wavelength shear waves ($v = \sqrt{G/\rho}$).

Physics. A lattice is mechanically stable only while $G_t > 0$, and the same lattice has a shear phonon whose frequency goes to zero there (Born criterion / soft-mode picture). Both curves fall and collapse together near failure; differences in between are real because G_t is the long-wavelength limit while the soft band averages many modes.

Real life. Defect-free crystals show a linear start, a peak of order $G/(2\pi)$ to $G/10$ (Frenkel) and shear-branch softening towards it. Real materials fail at a small fraction of that because of defects.

Assumes. Volume = $N_{\text{atoms}} \times \text{cell volume} / \text{atoms per cell}$; Morse α sets the peak; the stress drop at a break is a jump in this model (a real solid relaxes gradually), so G_t is not smooth right next to a break.

P6 Transport consequences

Why phonon softening matters for how fast carriers move in a transistor channel.

What you see. Three panels, all ratios to the unstrained crystal: (a) atomic mean-square displacement $\langle u^2 \rangle$ at 77, 300, 450 K; (b) phonon-limited mobility proxies (three temperatures plus G_t/G_0); (c) saturation-velocity and effective Debye-temperature proxies.

Data in. The full eigenvalue set for the stable range (`phononTransportMetrics.m`) and G_t/G_0 from P5 (your `centralDiffDerivative.m`).

How it is made.

- $\langle u^2 \rangle \propto (1/N) \sum (1/\omega_n) \coth(x_n/2)$, $x_n = \hbar\omega_n/(k_B T)$. The top unstrained mode is mapped to **15.5 THz**; modes below 5 % of the unstrained lowest frequency are floored so the sum cannot diverge.
- Mobility (Debye–Waller): $\mu/\mu_0 = \langle u^2 \rangle_0 / \langle u^2 \rangle$ (high-T phonon-limited resistivity $\propto \langle u^2 \rangle$). Acoustic-deformation-potential mobility: $\mu/\mu_0 = G_t/G_0$.
- Saturation velocity: $v_{\text{sat}}/v_{\text{sat},0} = \sqrt{(\omega_{\text{op}}/\omega_{\text{op},0})}$, with ω_{op} = mean of the top 10 % of modes (carriers stop accelerating once they can emit an optical phonon).
- Θ_D ratio from the logarithmic-mean frequency.

Physics. Softer phonons mean bigger atomic displacements (a more disordered potential for the electron), more thermally populated low-energy phonons to scatter from, and weaker elastic constants – all lowering phonon-limited mobility. The energy of the highest phonons separately sets the velocity at which carriers lose energy in bursts.

Real life. Strained-silicon mobility gains (tens of percent at ~1 % strain) come mostly from **band-structure** changes, which this model does not contain. Phonon frequencies in devices shift about 1 % per 1 % strain (Grüneisen parameter ~1), so the phonon effect is small at device strains and decisive only near mechanical failure. Expect $\langle u^2 \rangle$ flat then diverging, mobility proxy near 1 then falling steeply.

Careful with panel (c). In this spring model compressed Morse bonds stiffen, which can *raise* the top of the spectrum, so v_{sat} can go up. Do not present it as “softening lowers saturation velocity”.

Assumes. 15.5 THz reference, 77/300/450 K, the proportionality laws above (ratios only, no material constants), 5 % floor, top-10 % = “optical”, Gamma-point spectrum. These are scaling estimates, **not device predictions**.

4. Bond-breaking plots Q1–Q2

Q1 Bond map

Where in the crystal are bonds overstretched, and which have already snapped?

What you see. Three snapshots with every atom and spring projected on the xz plane. Colour = bond stretch (blue compressed, red stretched, symmetric range). Black dashed line with an $\times$ = broken bond. Titles give the number and % of broken bonds.

Snapshots. (1) first strain $\geq$ half the first-break strain, (2) the first-break strain, (3) the last stable strain. If nothing breaks, three evenly spaced strains.

Data in. Current bond vectors, stretch and “still present” flags from the sweep; the bond list `bonds(:,1)` (start atom) from **your** generator; atom positions = `pos +  $\gamma \cdot z$  + relaxation`. Bonds are drawn from least to most stretched so the critical ones lie on top.

Physics. For simple shear $u_x = \gamma z$, bonds along the **45° tension diagonal** stretch, the perpendicular diagonal compresses, and bonds along x or z barely change. Broken bonds line up in parallel rows – the slip planes.

Real life. A perfect crystal under simple shear does fail on the bonds most aligned with the tension axis and then slips as a band. Real crystals start at defects and surfaces, at smaller strain, so a pristine periodic result is an **upper bound** on strength.

Assumes. 17.3 % critical stretch for every spring; permanent breaking (no re-forming); every spring in the network can break, including the long-range springs the “auto rigid” builder adds for rigidity (Al_2O_3 uses 4 shells), so “bond” means “spring”; xz projection overlays bonds that differ only in y.

Q2 Break statistics

When do bonds break, and why those ones?

Left panel (when). % of bonds broken versus strain: black = all bonds, one coloured line per distinct bond length (rounded to 0.01). Steps mean a whole symmetry-equivalent family breaks at once.

Right panel (why). Each dot is a bond at the last stable strain: x = geometric factor $n_x n_z$ of its undeformed unit vector (`bonds(:,4:6)` from **your** list), y = its computed stretch. $\times$ = broken. Dashed line = first-order prediction $\varepsilon = \gamma n_x n_z$; dotted red line = critical stretch.

How it works. $|r + \gamma r_z \hat{x}|^2 = d_0^2(1 + 2\gamma n_x n_z + \gamma^2 n_z^2)$, so to first order the stretch is $\gamma n_x n_z$. Bonds with the largest positive factor reach the critical stretch first; scatter around the line is second-order and relaxation effects.

Physics. Bond breaking under shear is a geometric selection – which bonds go first depends on orientation, not position, unless a defect or surface concentrates stress. Comparing bond lengths shows whether short strong or long weak springs fail first.

Real life. Same ordering in any crystal for the affine part; molecular-dynamics studies of ideal shear also show 45° bonds failing first, followed by collective slip. Defects lower the strain and localise the first break.

Assumes. As Q1, plus the first-order line is valid only for small γ .

5. Assumptions at a glance

Plot	From your .m files	New computed data	Assumed / chosen
P1	<code>buildDynamicalMatrix</code> logic (validated), <code>autoRigidShells</code> , <code>parseCIFFile</code>	λ , soft band, event strains	Morse $\alpha = 4$, soft band = lowest 5 %, pre-stress, relaxation
P2	same	eigenvalues at $\gamma = 0$ and last stable	Gaussian width 1.5 %, Gamma-point only
P3	same	eigenvalues at every stable step	as P2
P4	<code>extractModeShape</code> , <code>generateLatticeGeneral</code> positions	lowest eigenvector, relaxation displacement	xz projection; one (maybe degenerate) mode
P5	<code>centralDiffDerivative</code> , virial as in <code>computeAtomicStress</code> (checked vs <code>computeShearForces</code>)	τ , G_t	volume per atom; Morse α sets the peak; smoothing of $\tau(\gamma)$
P6	(via P5) <code>centralDiffDerivative</code>	$\langle u^2 \rangle$, mobility, v_{sat} , Θ_D ratios	15.5 THz, 77/300/450 K, scaling laws, 5 % floor, top 10 % = optical
Q1	<code>bonds</code> list, positions	bond vectors, stretch, present-flags	17.3 % critical stretch, permanent breaking, all springs breakable
Q2	<code>bonds(:,3)</code> lengths, <code>bonds(:,4:6)</code> directions	stretch, broken fractions	first-order geometric law, length grouping tolerance 0.01

Global to every plot: one k for all springs (your model); central forces only (no bond-angle terms); shear in the xz plane; Gamma-point ($q = 0$) modes of the chosen supercell; periodic, defect-free lattice; reduced units.

Validation printed by `runPhononShearAnalysis.m` (Al_2O_3): affine model vs `applyShearForce.m` frequencies: 7×10^{-16} ; affine stress vs `computeShearForces.m` virial: identical; $\gamma = 0$ spectrum vs `buildDynamicalMatrix.m`: 10^{-15} ; relaxation residual: below 10^{-9} .

6. Model vs real crystals

Topic	This model	Real crystal
Forces	central springs, one k ; Morse at large strain	bond-bending terms, different strengths per bond type, long-range Coulomb in ionic solids
Frequency scale	reduced units $\sqrt{k/m}$	THz / cm^{-1} from real force constants
Softening at a few % shear	small, quadratic in γ	same order: sub-percent shifts, mostly mode splitting

Ideal shear strength	set by α ; about γ 0.2–0.3 here	0.1–0.3 for defect-free crystals; real failure far earlier
Failure	all equivalent bonds break together	starts at defects/surfaces and spreads gradually
Instability	first mode with $\omega^2 = 0$ that fits the supercell	a soft branch at a specific wave-vector (may not fit a small cell: Al_2O_3 goes unstable near γ 0.31 for $N = 1$ but near 0.17 for $N = 2$)
Electron transport	phonon-side scaling only	dominated by band structure at device strains
Beyond the instability	plots stop	a new, transformed structure

7. Reading a result honestly – checklist

1. Did the soft band fall smoothly *before* the first bond broke? That is genuine geometric/anharmonic softening.
2. Did $\omega^2 < 0$ come before or after the first break? (CaF_2 : before; Al_2O_3 with $N = 1$: after.) Before = a pure elastic instability; after = rupture triggers the failure.
3. Change N . If the instability strain drops a lot, the soft mode has a wavelength the small cell could not hold.
4. Compare with the affine (blue) curve: the difference is the effect of relaxation, pre-stress and anharmonicity, not of your slider model.
5. Treat P6 as scaling ratios – “phonon contribution to the trend”, never absolute mobility.
6. If the crystal is barely rigid at $\gamma = 0$ (tiny lowest eigenvalue), trust the soft band over the single lowest mode.

Generated for the Version 7 project. Companion file: `Version7_plot_log.md`. Run the analysis from the app (Shear Analysis window) or with `runPhononShearAnalysis.m`.